ELEC361 MATLAB PROJECT2

FM Modulation and Demodulation

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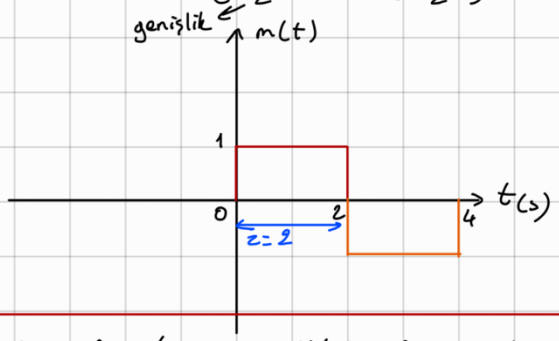
1 Analytic solutions

FM Modulation and Demodulation

The message signal $m(t) = \prod\left(\frac{t-1}{2}\right) - \prod(\frac{t-3}{2})$, t in seconds, modulates the carrier signal $c(t) = 5Cos(500\pi t)$ to obtain FM signal $y(t) = 5Cos(500\pi t + \emptyset(t))$, where $\emptyset(t) = 2\pi k_f \int m(\tau)d\tau$ with $k_f = 50$. The demodulator consists of a differentiator, a diode (assume ideal), a low pass filter (LPF) and a capacitor for DC blocking, successively.

1) Plot the message signal. Obtain and plot the magnitude spectrum of the message signal.
2) Obtain and plot the time-domain representation and magnitude spectrum of the phase $\emptyset(t)$.
3) Obtain and plot the time-domain representation and magnitude spectrum of the modulated signal $y(t)$.
4) Obtain and plot the time-domain representation and magnitude spectrum of the signal at each step of the demodulator. (Do the filtering in frequency domain.) Comment on the signal at each step. Compare the message signal and demodulated signal, and comment on them.
5) Do the steps above that can be done analytically, and compare to the simulation results and comment on them.

1) $m(t) = \Pi\left(\frac{t-1}{2}\right) - \Pi\left(\frac{t-3}{2}\right)$



$x(t) = A\Pi(t/z) \Rightarrow X(f) = Az\,sinc(fz)$

$\Pi\left(\frac{t}{2}\right) \overset{F}{\leftrightarrow} 2\,sinc(2f)$

Zamanda ötteleme özelliği (Timeshifting)

$x(t) \overset{F}{\leftrightarrow} X(f)$

$x(t-t_0) \overset{F}{\leftrightarrow} X(f)e^{-j2\pi f t_0}$

$\Pi\left(\frac{t-1}{2}\right) \overset{F}{\leftrightarrow} 2\,sinc(2f)e^{-j2\pi f}$

$-\Pi\left(\frac{t-3}{2}\right) \overset{F}{\leftrightarrow} -2\,sinc(2f)e^{-j2\pi 3f}$

$m(t) \overset{F}{\leftrightarrow} M(f) = 2\,sinc(2f)\left(e^{-j2\pi f} - e^{-j2\pi 3f}\right)$



$sinc(x) = \frac{\sin(\pi x)}{\pi x}$

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$e^{jx} = \cos x + j\sin x$

$\hookrightarrow M(f) = 2\,\mathrm{sinc}(2f)\left[\cos(\pi f) - j\sin(\pi f) - \cos(3\pi f) + j\sin(3\pi f)\right]$

$\Rightarrow |M(f)| = \sqrt{\left[\cos(\pi f) - \cos(3\pi f)\right]^2 + \left[-\sin(\pi f) + \sin(3\pi f)\right]^2}\ \ |2\,\mathrm{sinc}(2f)|$

so $|M(0)| = 0$, $|M(1{,}24)| \approx 0{,}51$

b)

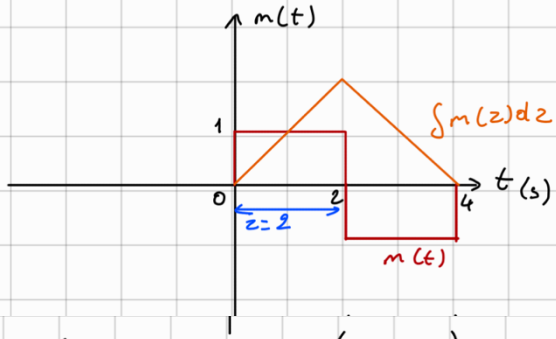
$$\int_{-\infty}^{t} m(z)\,dz = \begin{cases} \int_{-\infty}^{t} 1\,dz = t & t \le 2 \\ \int_{-\infty}^{2} 1\,dz + \int_{2}^{t} -1\,dz \\ 2 - t + 2 = 4 - t & 2 < t \le 4 \end{cases}$$

$\phi(t) = 2\pi k_f \int m(z)\,dz$

when $k_f = 50$

$\phi(t) = 100\pi \int m(z)\,dz$

$m(t) = \Pi\left(\frac{t-1}{2}\right) - \Pi\left(\frac{t-3}{2}\right)$



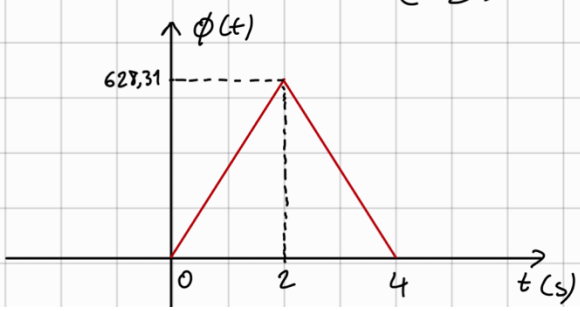
$\Rightarrow \int m(z)\,dz = 2 \cdot \Lambda\left(\frac{t-2}{2}\right)$

$2z = 4 \quad z = 2$

$\Rightarrow \phi(t) = 100\pi \cdot 2\,\Lambda\left(\frac{t-2}{2}\right)$

$= 200\pi\,\Lambda\left(\frac{t-2}{2}\right)$

$\approx 628{,}31\,\Lambda\left(\frac{t-2}{2}\right)$



$A\,\Lambda\left(t/z\right) \leftrightarrow A z\,\mathrm{sinc}^2(fz)$

Zamanda öteleme özelliği (Timeshifting)

$x(t) \xleftrightarrow{F} X(f)$

$x(t - t_0) \xleftrightarrow{F} X(f)\,e^{-j2\pi f t_0}$

$\Lambda\left(\frac{t}{2}\right) \xleftrightarrow{F} 2\,\mathrm{sinc}^2(2f)$

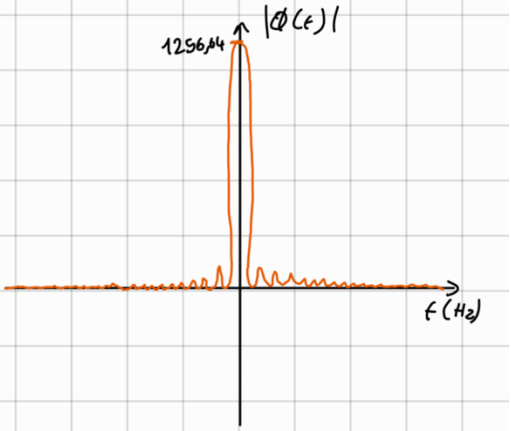
$\Lambda\left(\frac{t-2}{2}\right) \xleftrightarrow{F} 2\,\mathrm{sinc}^2(2f)e^{-j2\pi f 2}$

$\phi(F) = 200\pi \cdot 2\,\mathrm{sinc}^2(2f)e^{-j2\pi f 2}$

$= 400\pi\,\mathrm{sinc}^2(2f)\,e^{-j2\pi f 2}$

$\approx 1256{,}64\,\mathrm{sinc}^2(2f)e^{-j2\pi f 2}$

$|\phi(f)| = 1256{,}64\,\mathrm{sinc}^2(2f)$



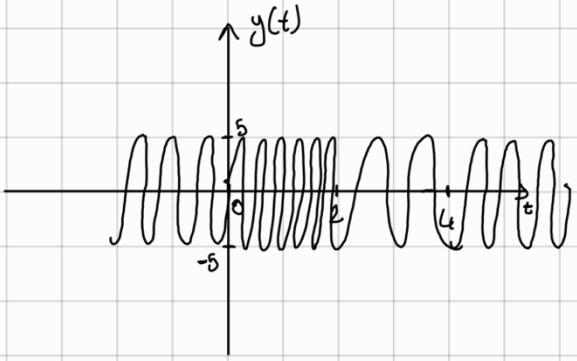
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3) $y(t) = 5\cos(500\pi t + \phi(t)) \Rightarrow$

$$y(t) = \begin{cases} 5\cos(500\pi t + 100\pi t) = 5\cos(2\pi 300 t) & 0 < t < 2 \\ 5\cos(500\pi t + 100\pi(4-t)) = 5\cos(2\pi 200 t + 400\pi) & 2 < t < 4 \\ \phi(t) = 0 \Rightarrow y(t) = 5\cos(2\pi 250 t) & \text{other} \end{cases}$$

(frequency 350; frequency of 200; frequency of 250)

So it can be said that $Y(t)$ will have dominant components at these frequencies.



$\Delta f = k_f m_p = 50 \cdot 1 = 50$

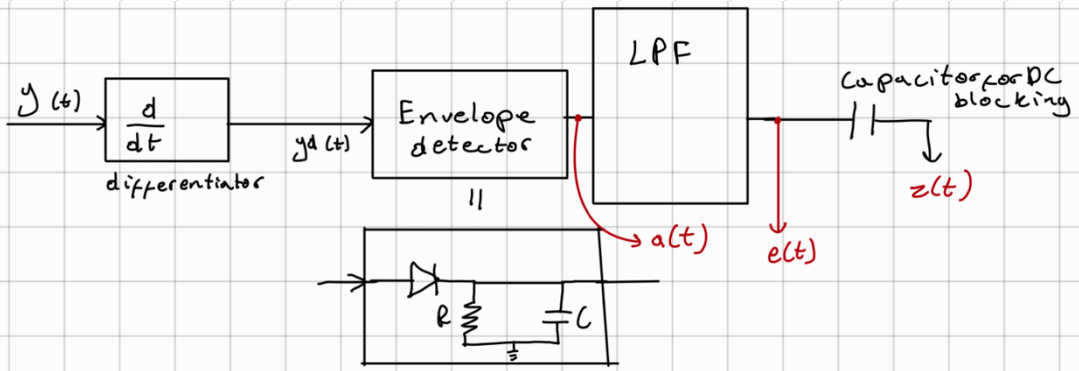
$\beta = \frac{\Delta f}{B} = \frac{50}{B}$

$BG = 2B(\beta + 1)$

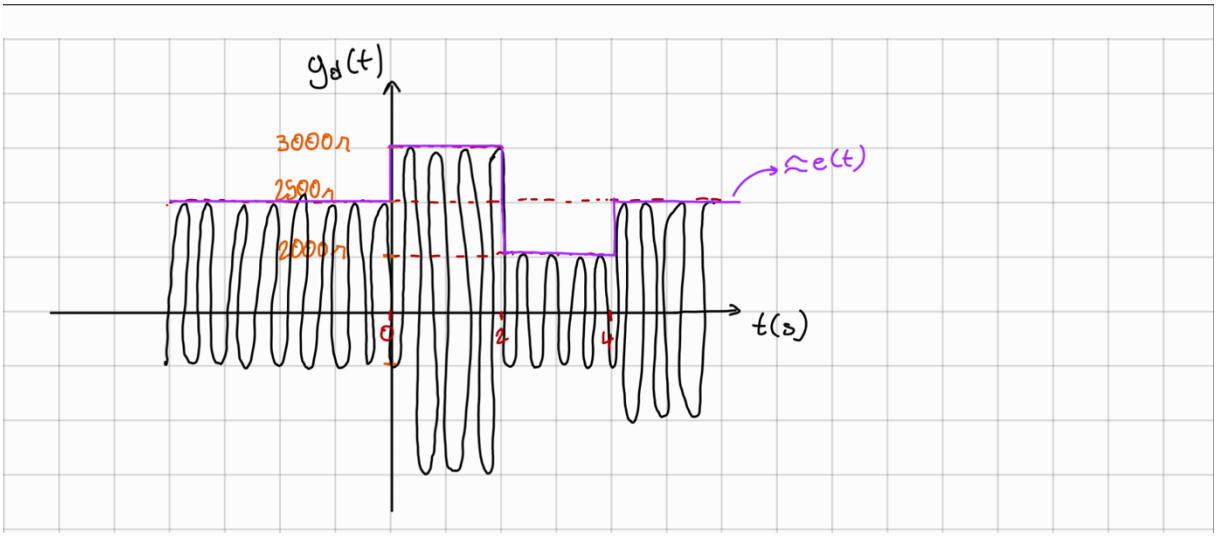
$= 2(50 + 2B)$

$= (100 + 2B)$ Hz //

4)



$$y_d(t) = \frac{dy(t)}{dt} = \begin{cases} \frac{d}{dt} 5\cos(2\pi 300 t) & = -3000\pi \sin(2\pi 300 t) & 0 < t < 2 \\ \frac{d}{dt} 5\cos(2\pi 200 t + 400\pi) & = -2000\pi \sin(2\pi 200 t + 400\pi) & 2 < t < 4 \\ \frac{d}{dt} 5\cos(2\pi 250 t) & = -2500\pi \sin(2\pi 250 t) & \text{other} \end{cases}$$


$g_d(t)$
3000π
2500π
2000π
≃e(t)
0
2
4
t(s)

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2 MATLAB solutions

2.1 Plot of m(t) signal. M(f) signal and the plot of the magnitude spectrum.

m(t) signal Plotted in MATLAB in time domain as shown in Figure 1 and by using FFT the magnitude spectrum of m(t) obtained and plotted in Figure2.

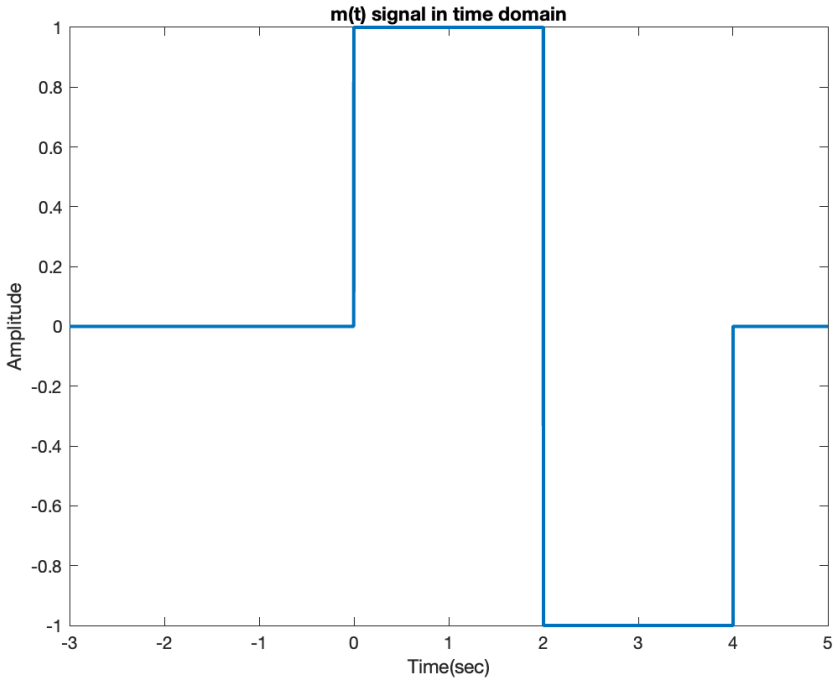


Figure 1: m(t) signal plot in time domain.

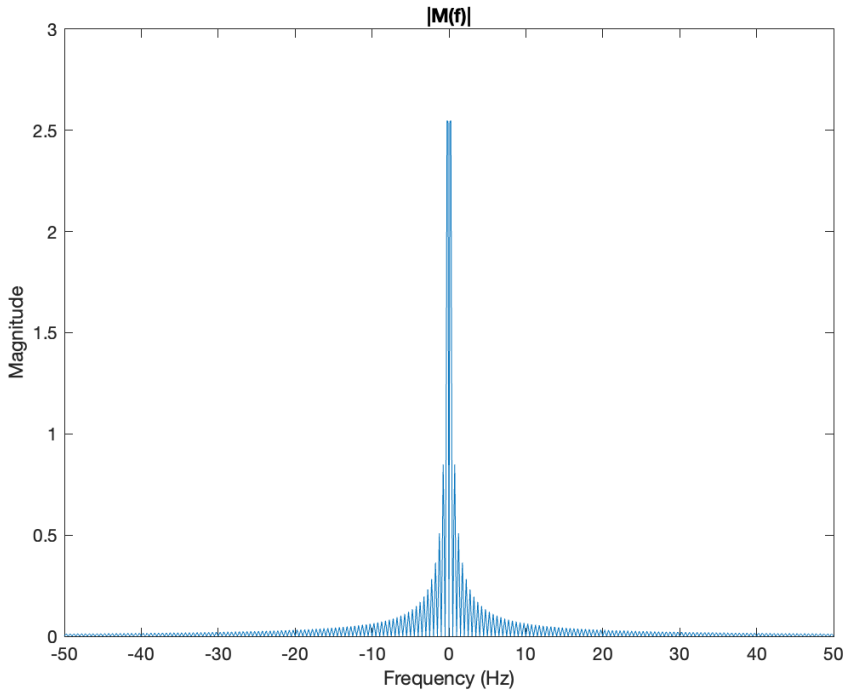


Figure 2: |M(f)| plot the magnitude spectrum of m(t) signal.

2.2 Obtain fi(t) and plot it, and their magnitude spectrum.

By integrating the signal m(t) in section 2.1, and scale it with 2*pi*kf = 100 pi the signal fi(t) obtained and plotted in Figure 3. Using FFT the magnitude spectrum of fi(t), |Fi(f)| is obtained and plotted in Figure4. To integrate m(t) signal Least squares integration method is used to integrate faster the integral command of MATLAB. And it's okay because the m(t) signal is simple.

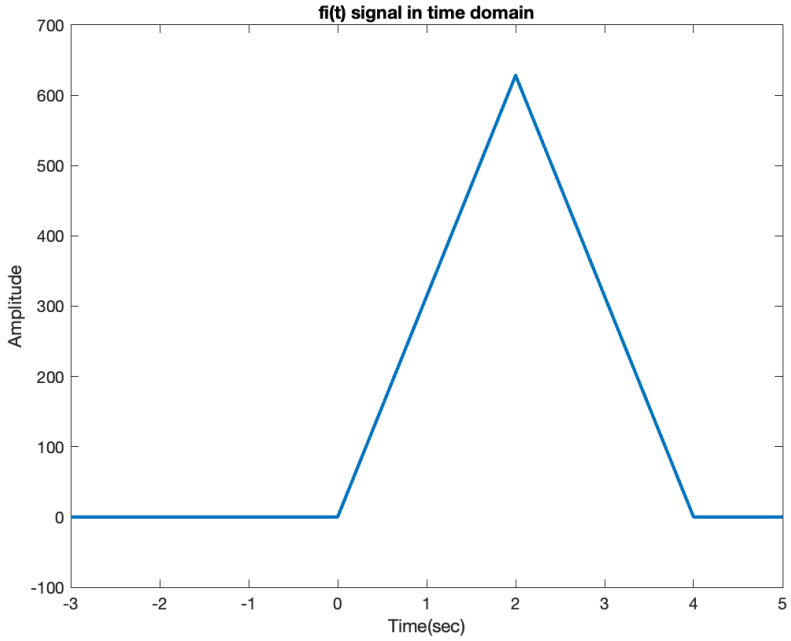


Figure 3: fi(t) signal plot in time domain.

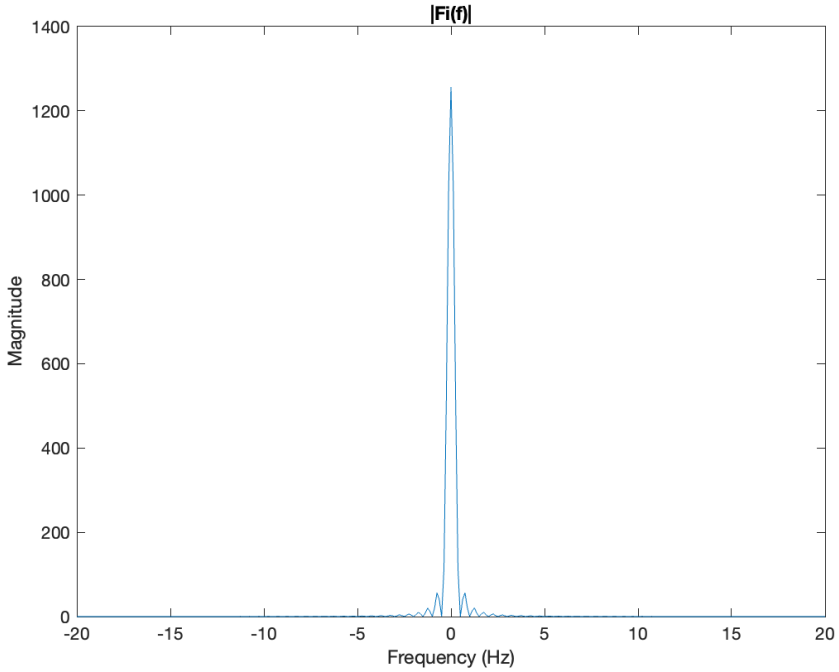


Figure 4: |Fi(f)| plot the fi(t) spectrum magnitude plot.

2.3 y(t) signal in time domain and |Y(f)| the magnitude spectrum of y(t).

By putting the phase fi(t) inside the carrier, as 5*cos(2*250*pi + fi(t)) the y(t) signal is obtained and plotted in Figure 5.

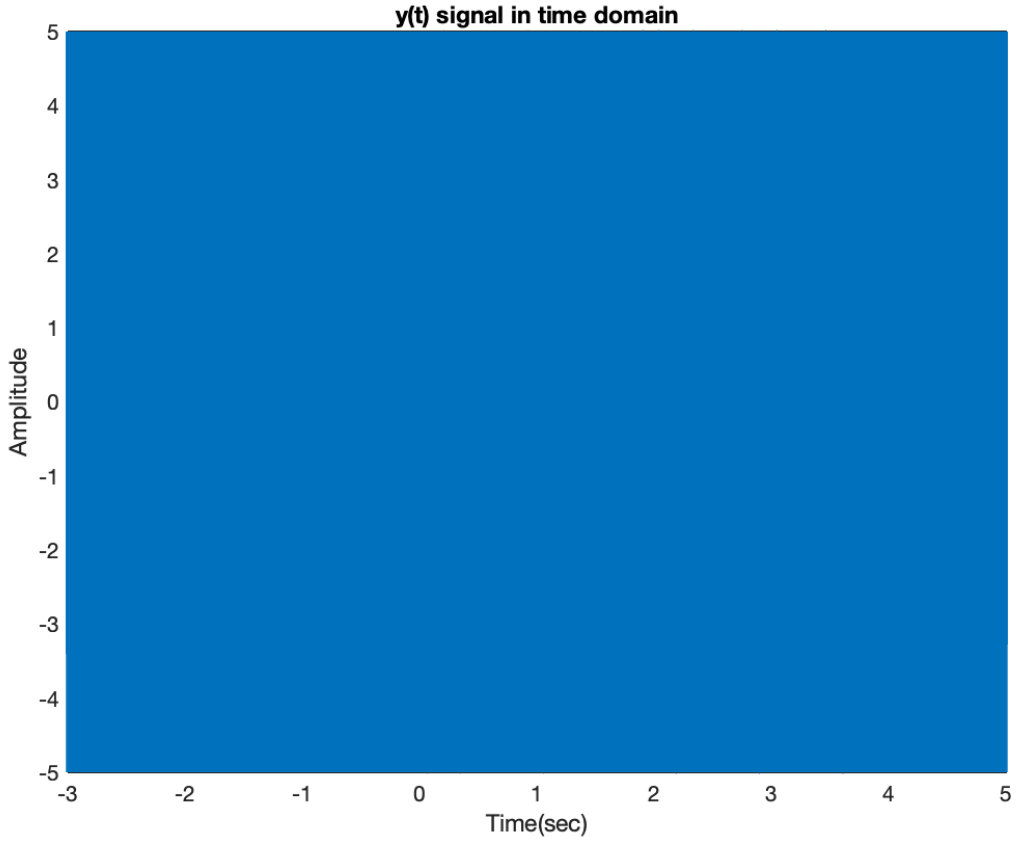


Figure 5: y(t) signal plot in time domain

When y(t) plotted between 3.95 – 4.05 and 1.95 - 2.05 as shown in Figure 6 and Figure 7 , the change in frequency of the signal at the transition points is clearly visible as shown in the analytic section.

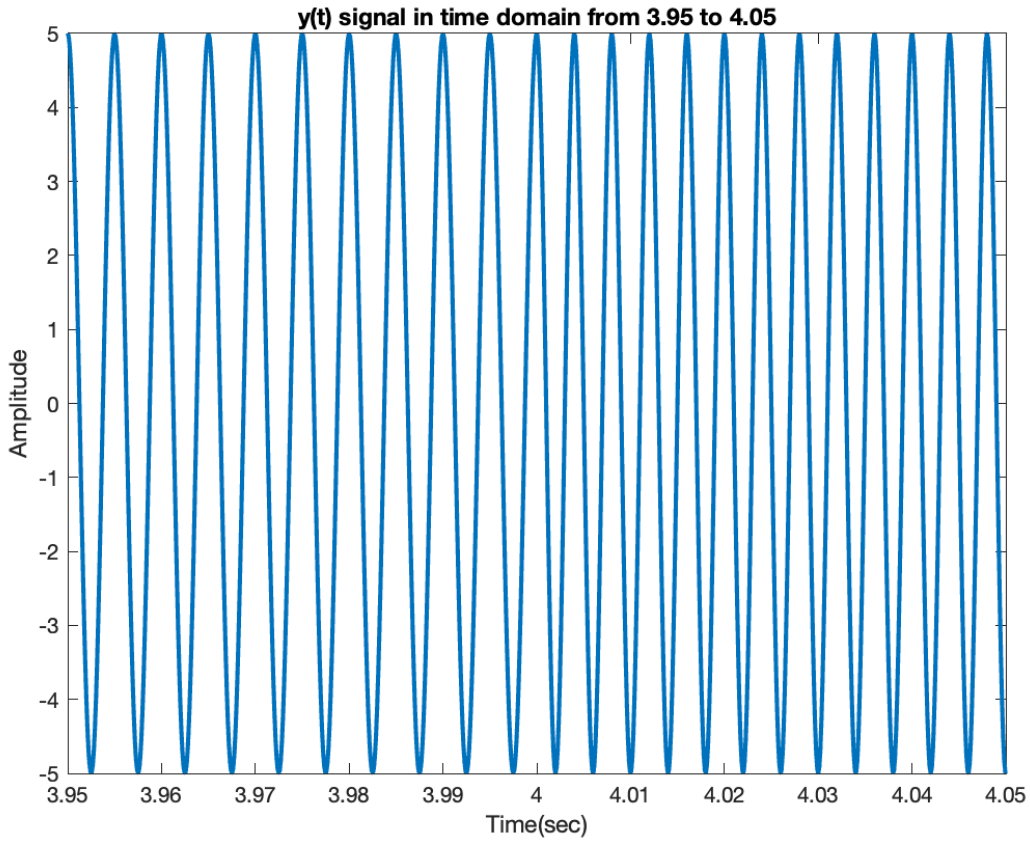


Figure 6: y(t) signal plot in time domain from 3.95 to 4.05

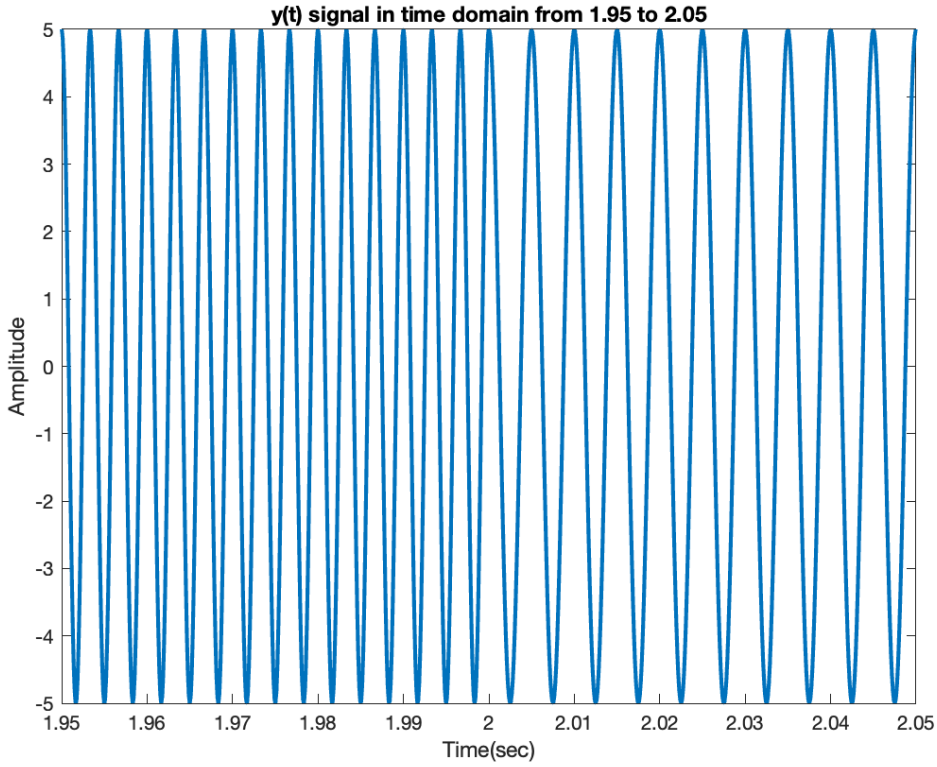


Figure 7: y(t) signal plot in time domain from 1.95 to 2.05

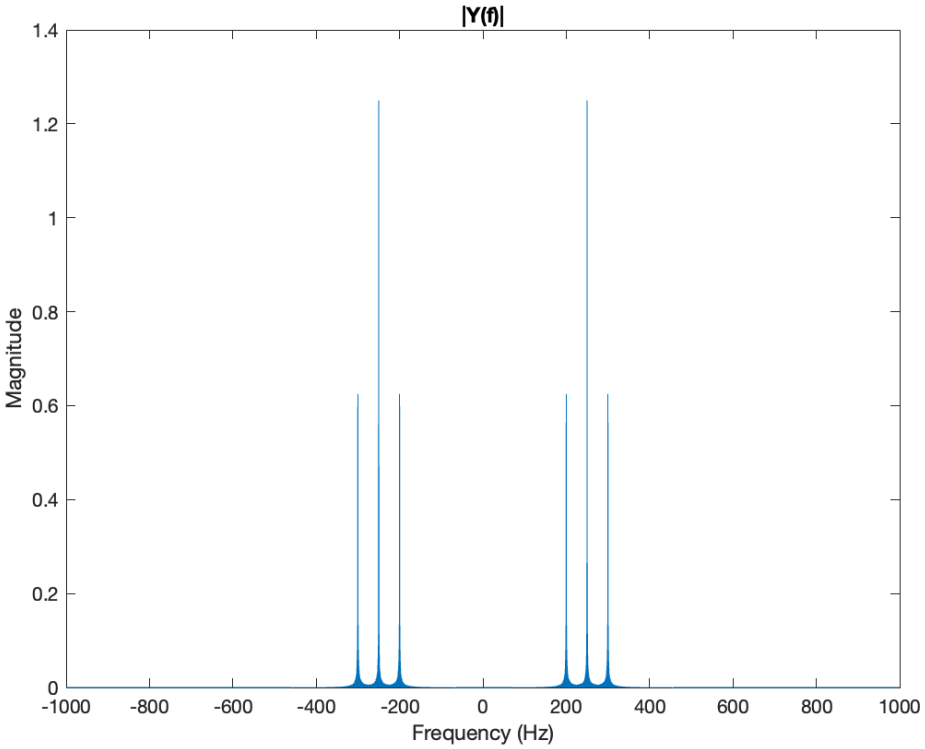


Figure 8: |Y(f)| the y(t) signal magnitude spectrum

The Magnitude spectrum of signal y(t) is obtained in Figure 8 and It is clearly seen that it has dominant components at frequencies 250, 200 and 300.

2.4 The time domain signals and their magnitude spectrum at each step of demodulator.

The derivative of the signal y(t) is obtained by using MATLAB diff function and it's called yd(t). The signal is plotted in time domain as shown in Figure 9. The magnitude spectrum of yd(t) is obtained using FFT and plotted in Figure 10.

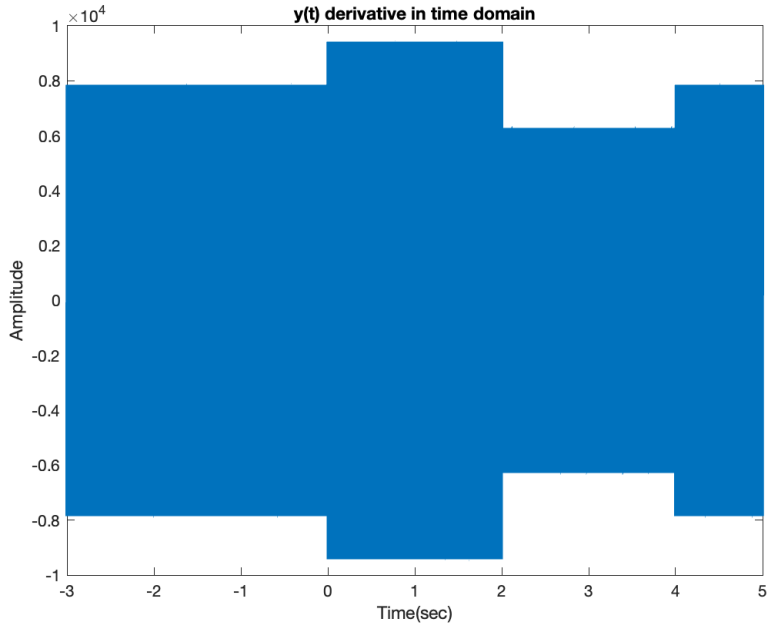


Figure 9: y(t) derivative plot in time domain

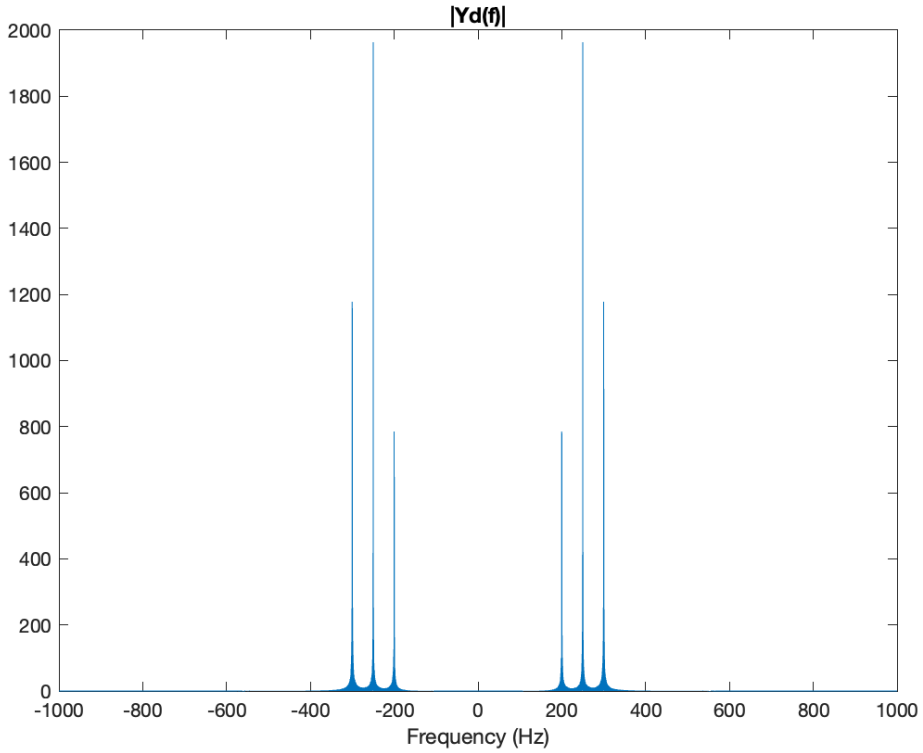


Figure 10: |Yd(f)| the y(t) derivative magnitude spectrum

The signal yd(t) passed by the envelops detector and a(t) signal obtained. Resistor and Capacitor values is chosen as R= 100kohm and C= 1uF. The a(t) signal is plotted in Figure 11 And the magnitude spectrum of a(t) is obtained and plotted in figure 12.

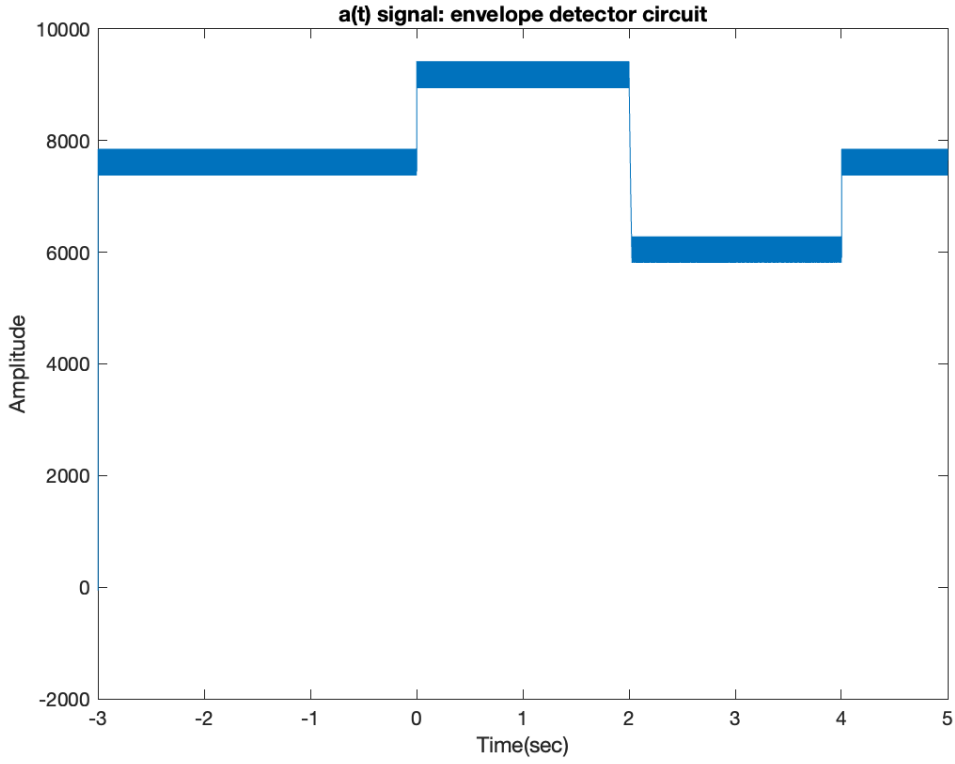


Figure 11: a(t) signal plot in time domain the output of the envelope detector

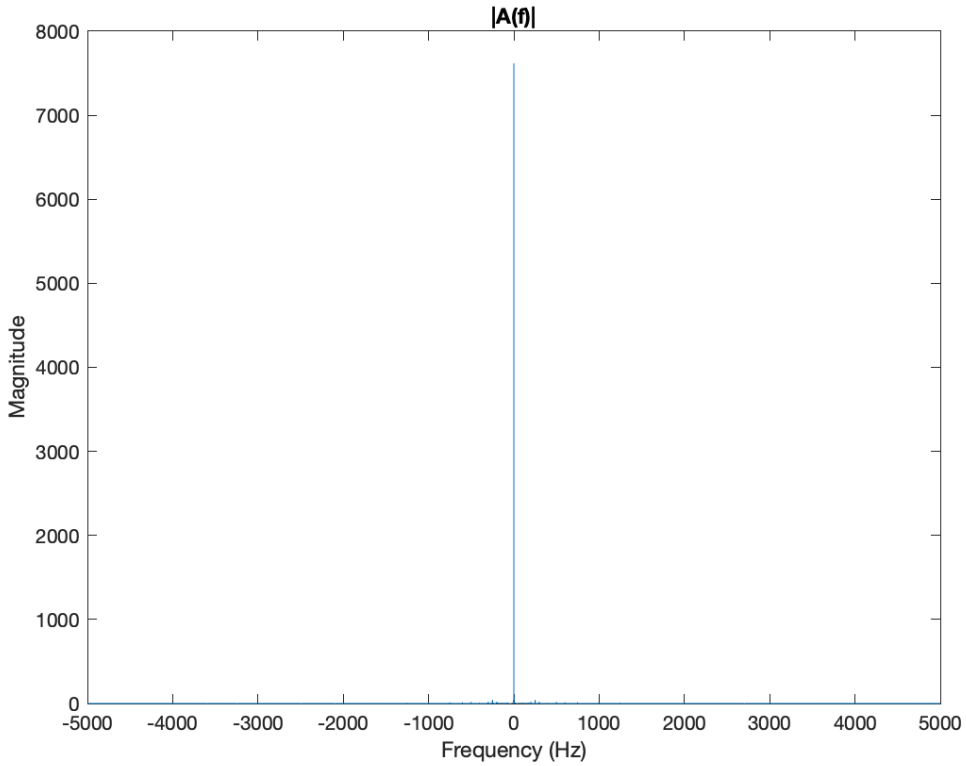


Figure 12: |A(f)| plot the magnitude spectrum of a(t)

To Filter the ripples at high frequency the signal a(t) is passed by a low pass filter and this signal is called e(t). First the spectrum of it is obtained after is passed by a filter in the frequency domain than the signal e(t) is obtained by using IFFT as shown in Figure 13 and Figure 14.

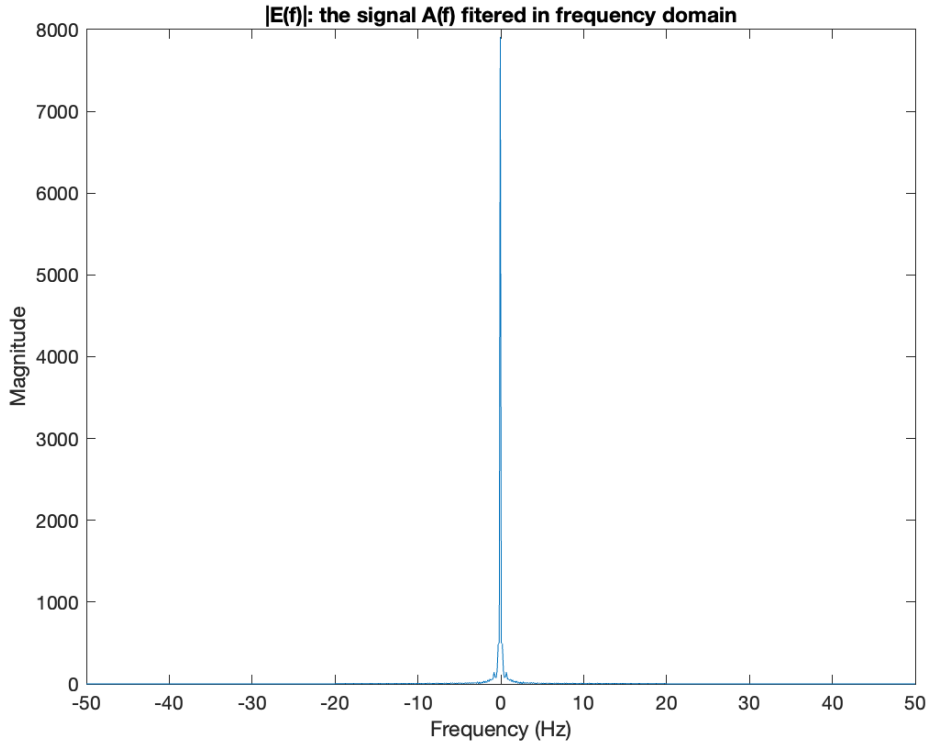


Figure 13:|E(f)|: the signal A(f) filtered in frequency domain

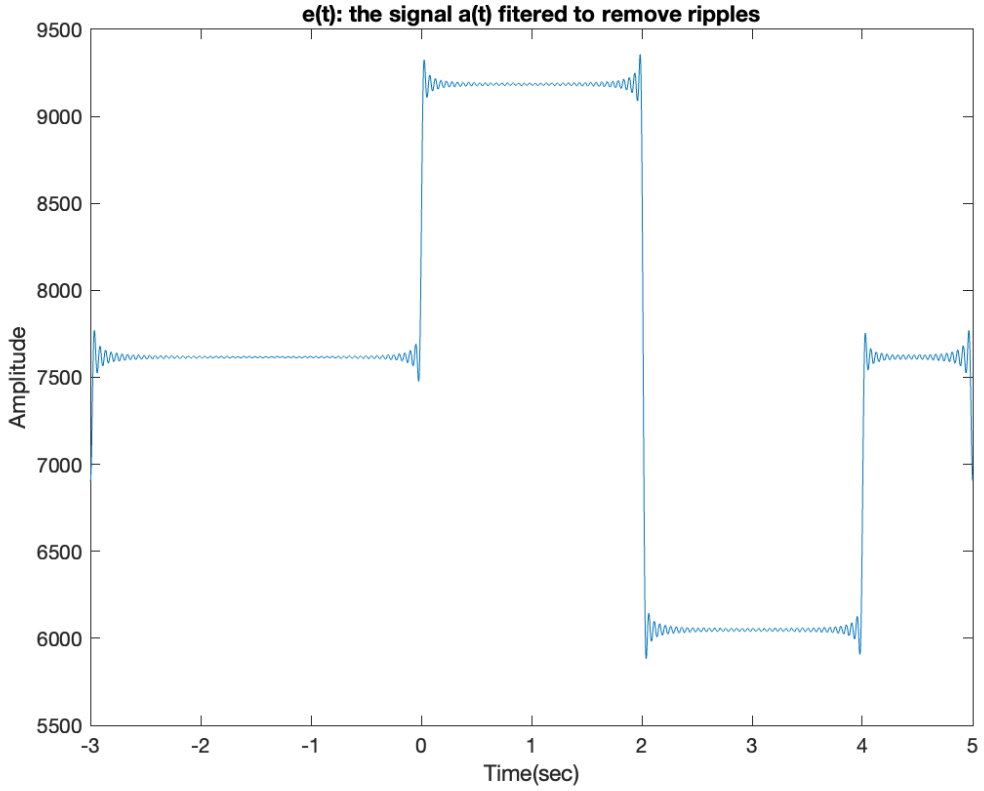


Figure 14:e(t) plot, the signal a(t) fitered to remove ripples

The signal e(t) is normalized by Ac*2*pi*kf to obtain the amplitude of the message as the same of m(t). en(t) signal is plotted as Figure 15.

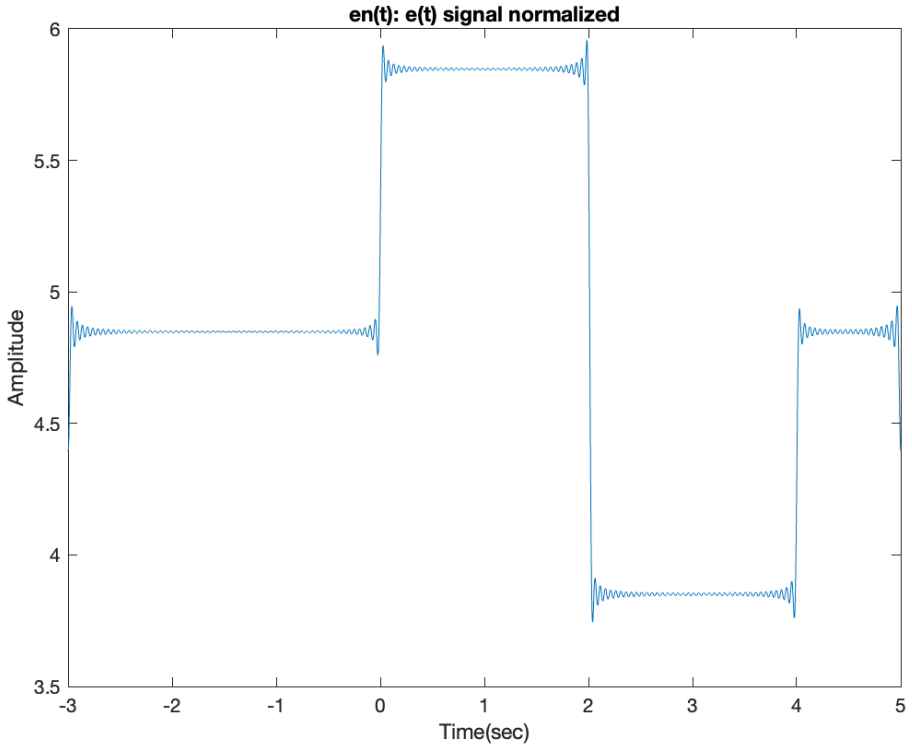


Figure 15: en(t) plot the e(t) signal normalized.

The m(t) and en(t) signals is plotted together as shown in Figure 16. A DC component is briefly seen in en(t) signal to ignore the DC component a Capacitor is used, and the signal passed by the DC blocking capacitor is called z(t) and it's plotted as shown as Figure 17. The magnitude spectrum of z(t) is obtained and plotted as shown in Figure 18.

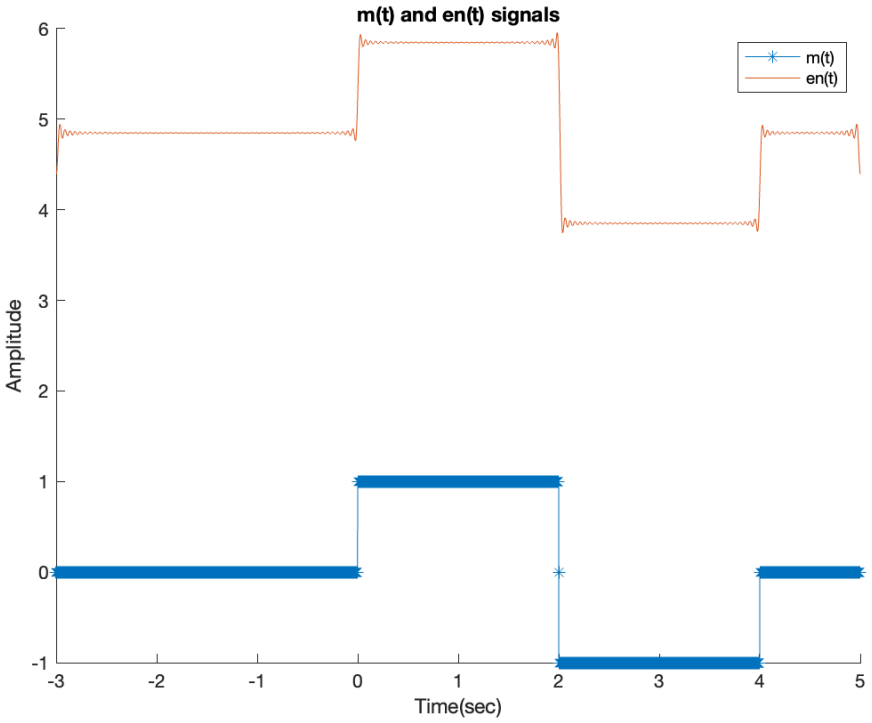


Figure 16: en(t) and m(t) signals plotted together.

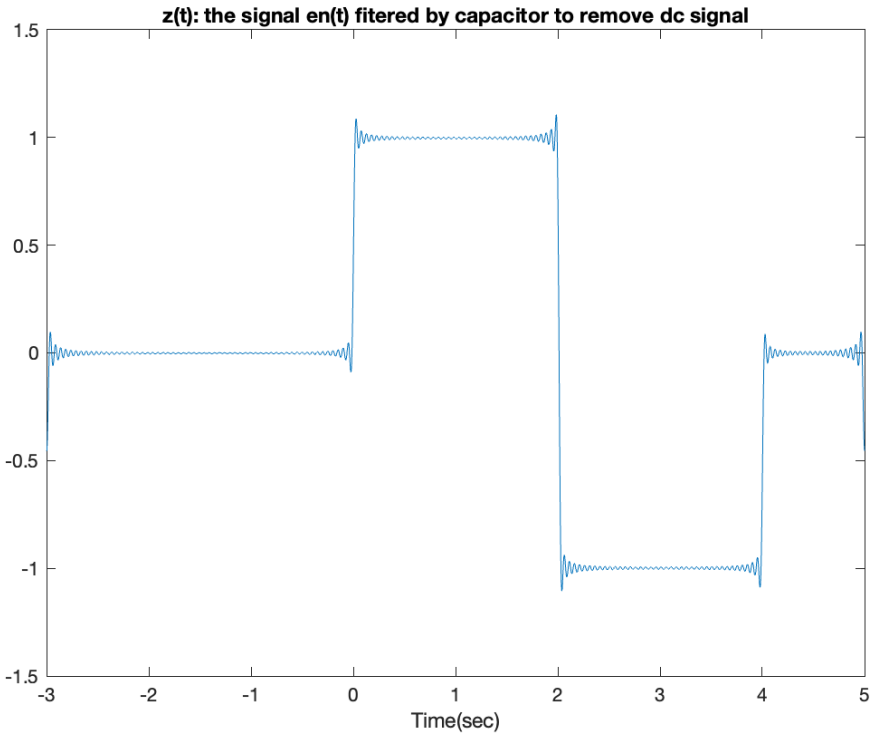


Figure 17: z(t) signal plot the signal en(t) after dc blocking capacitor

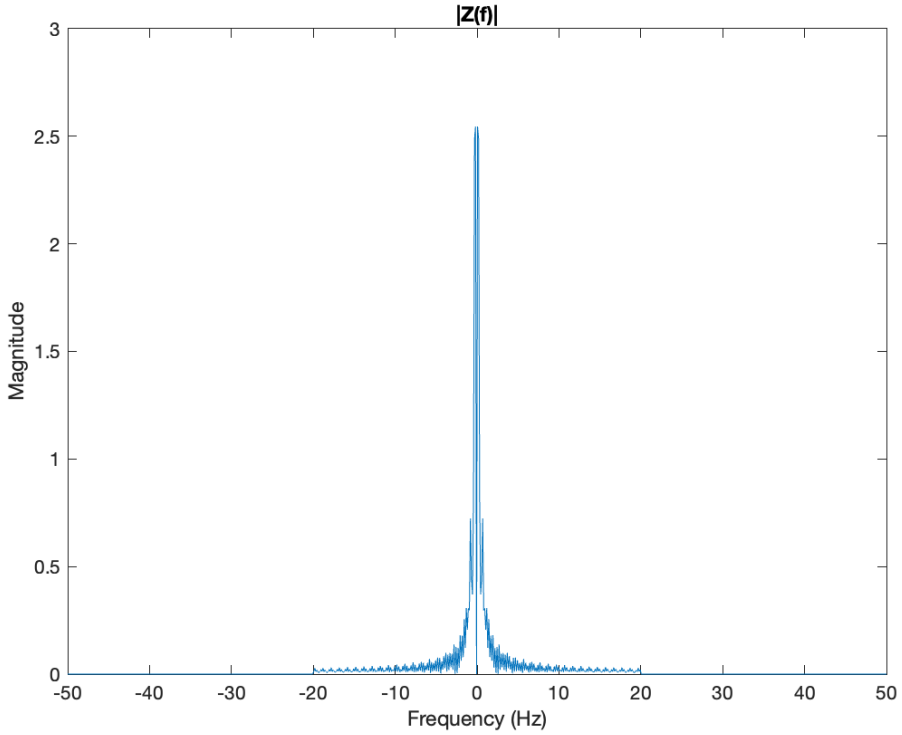


Figure 18: |Z(f)| plot

The m(t) and z(t) plotted together as shown in figure 19 and they appear to overlap. But z(t) signal have some noise at the transition points.

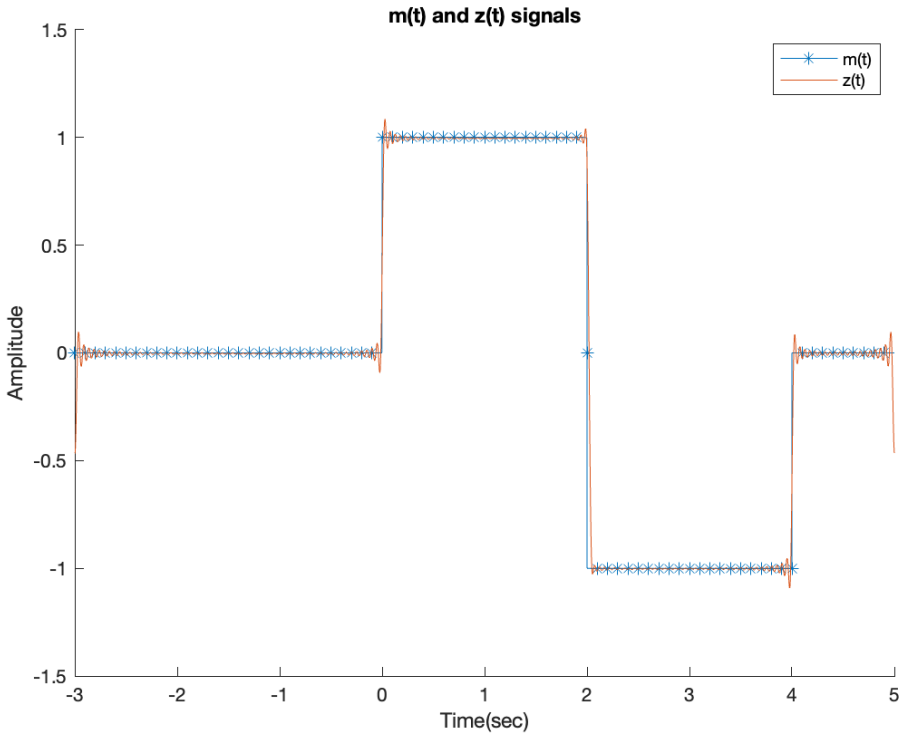
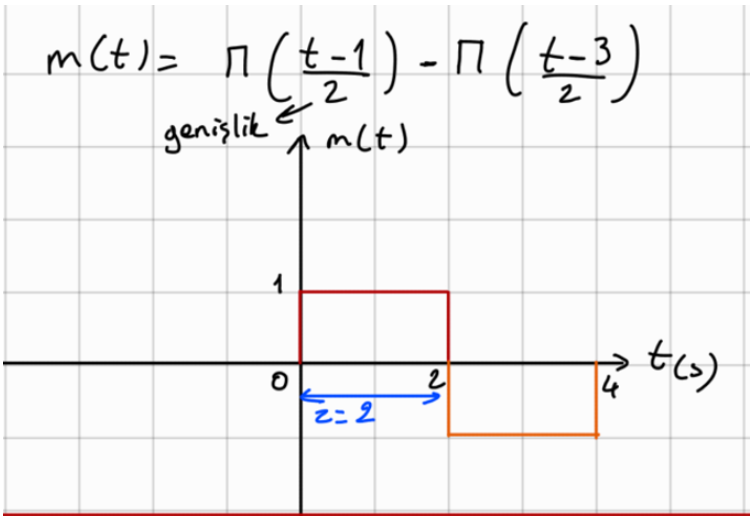


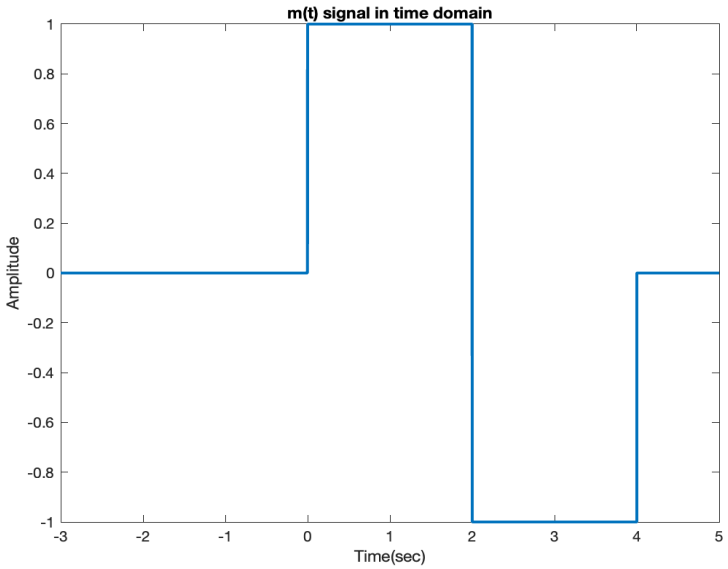
Figure 19: m(t) and z(t) signals plotted together.

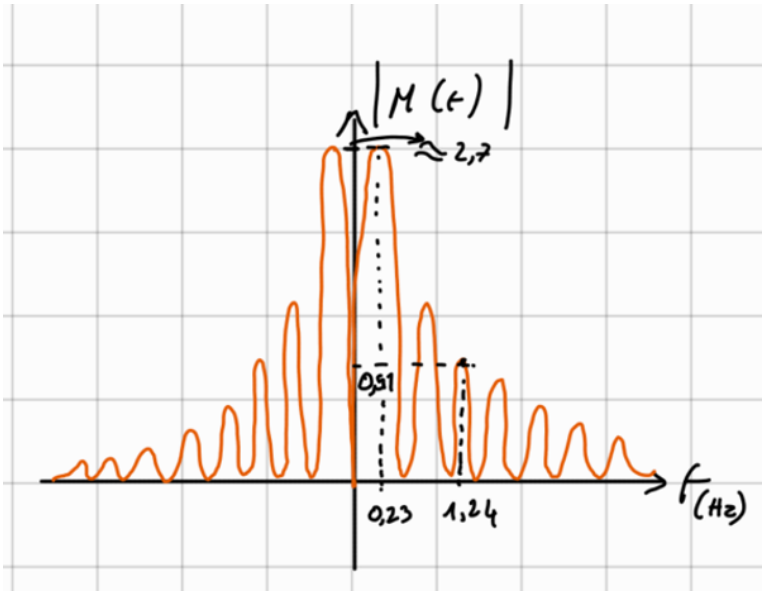
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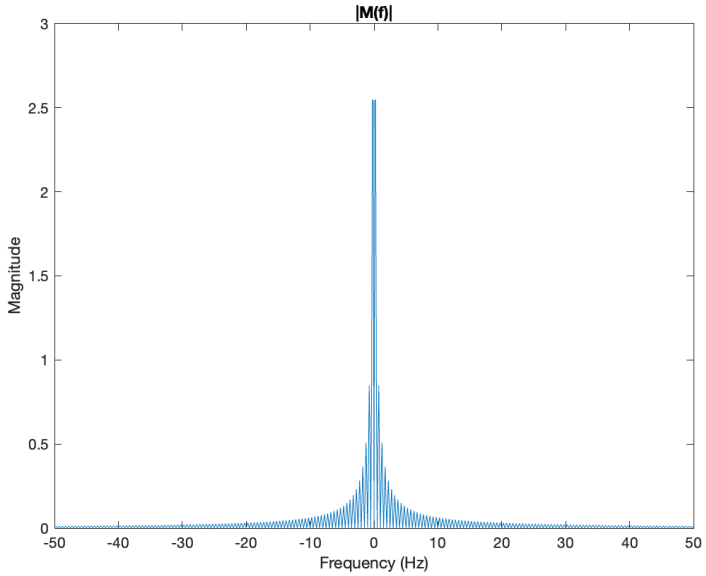
3 MATLAB VS Analytic solution

As shown below m(t) signal and |M(f)| obtained in the analytic and MATLAB and results are consistent.

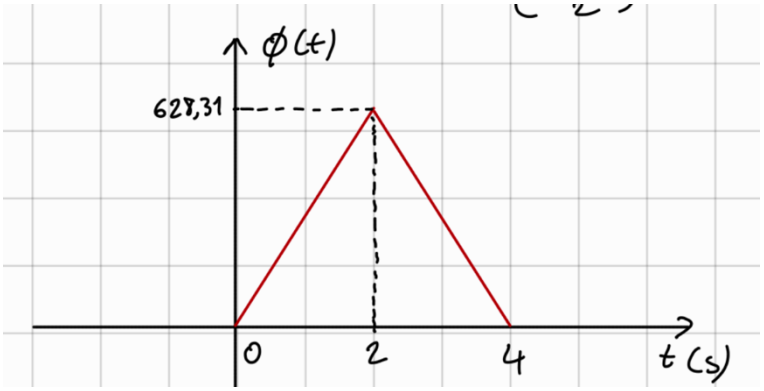


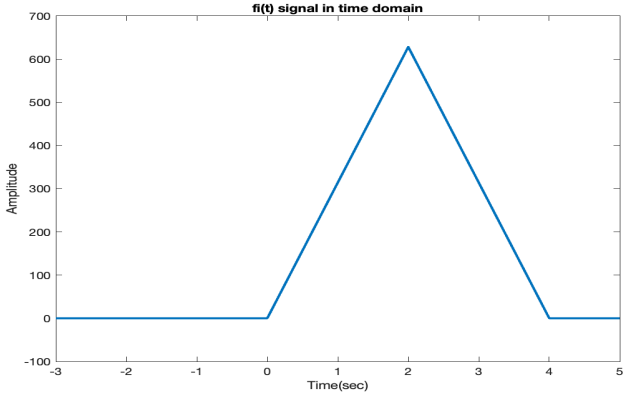




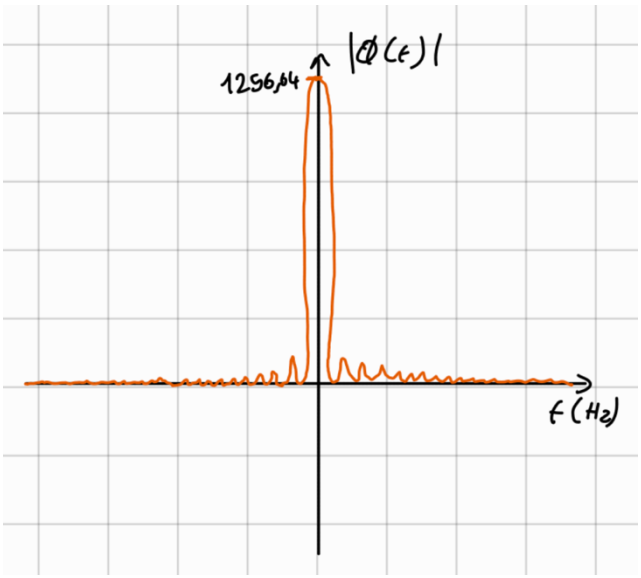


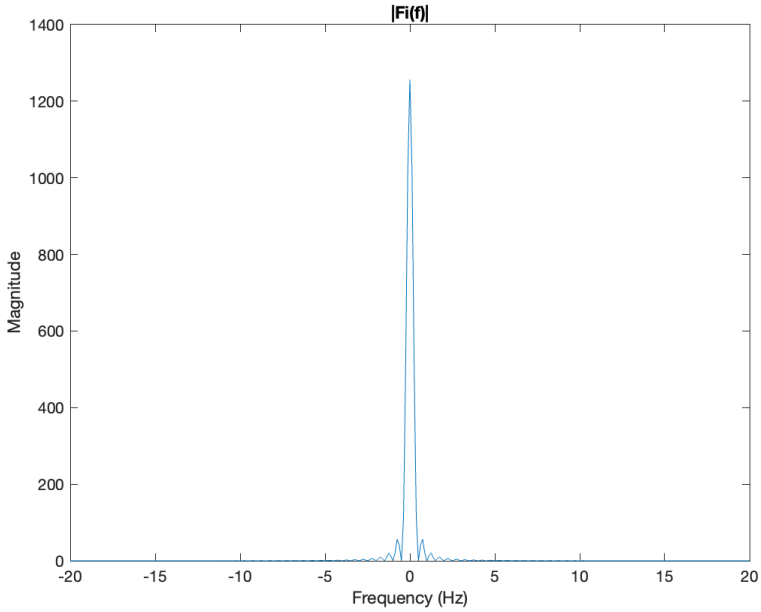
Then fi(t) and |Fi(f)| obtained as below by integrating m(t) and scale it with 2*pi*kf. And results are consistent.



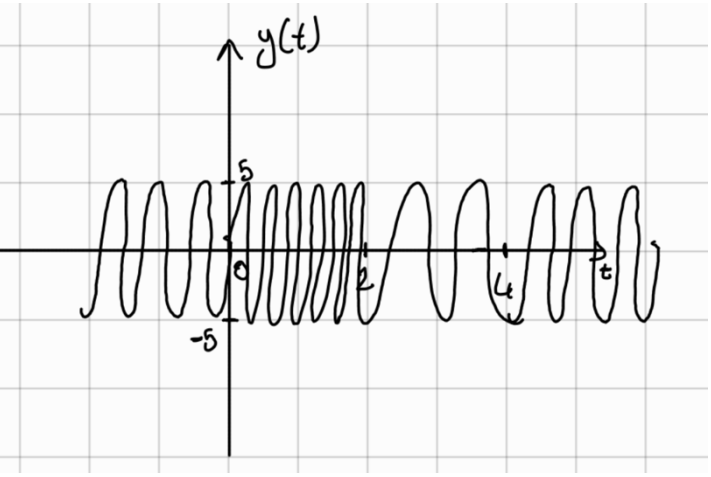


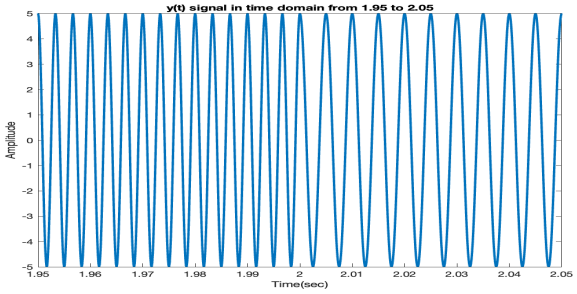
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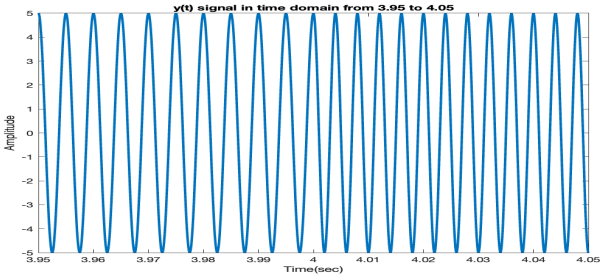




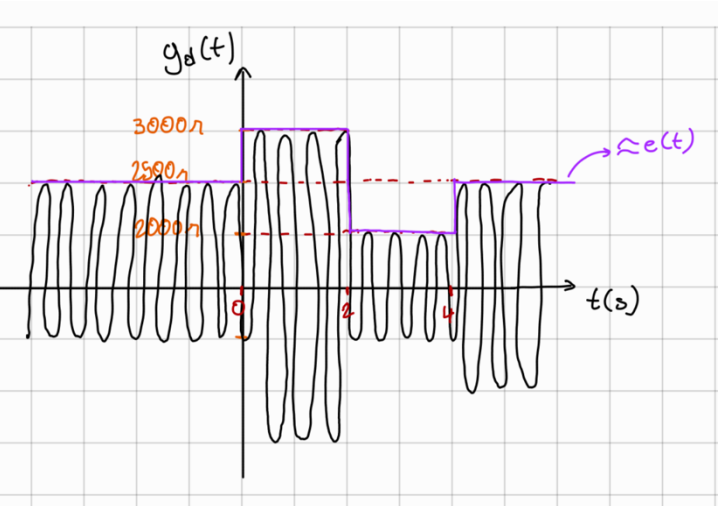
y(t) signal and its magnitude spectrum are obtained and plotted. And results are consistent in Analytic and MATLAB. (The y(t) signal is drawn notationally in analytics to clearly show the frequency change at transition points.)

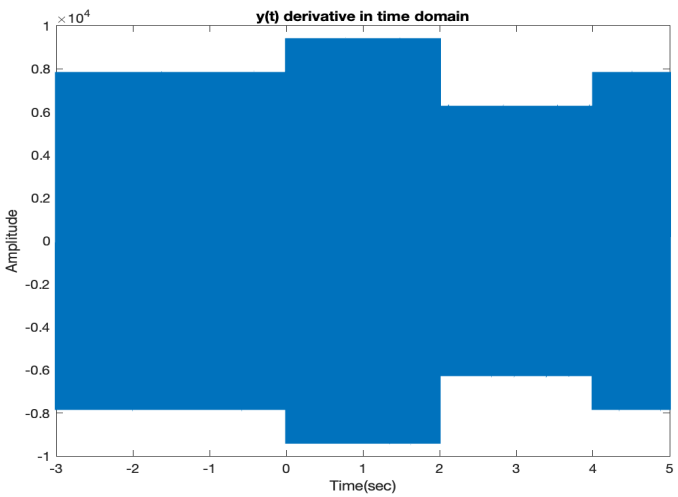






The derivative of y(t) is obtained as following the amplitude founded in analytic between 0 and 2 is 3000*pi = 9424.78 = 0.942478 * 10^-4 and it's the same as found in MATLAB.





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In this section, the comparison of important signals is mentioned, and other signals are given in more detail in the 1st and 2nd sections.

4 MATLAB CODE

```
save_path = 'figures/';
mkdir(save_path)
%%
fig_counter = 1;
%%
Ts = 0.00001; %determination of the sample coefficient
%define the time vector for simulating the signal in time
Tlow = -3;
time = Tlow:Ts:5-Ts;
%define the m(t) signal
m = @(t) square_wave((t - 1)./2 ) - square_wave((t - 3)./2 );
%%
%Calculations
mt = m(time);
%%
%plot m(t) in the time domian
figure Name 'm(t) in time domain'
grid on;
plot(time, mt,"LineWidth",2)
title('m(t) signal in time domain')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- m(t) in time
domain'),'png')
fig_counter=fig_counter+ 1;
%%
%%Take fourier transform using fft
fs = 1/Ts;
mf = fft(mt)/fs;%take the fourier transform using fft
figure Name 'M(f)'
n = length(mt);
%take the shift of the spectra of the fourier transform
fshift = (-n/2:n/2-1)*(fs/n);
mfshift = fftshift(mf);
plot(fshift,abs(mfshift))
xlim([-50,50])%limit the frequency x axis
% between -50 and 50 for best view
title('|M(f)|')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- M(f)'),'png')
fig_counter=fig_counter+ 1;

%%
kf = 50;
%fi = @(t) 2*pi*kf.*integral(m,-1,t);

fit = zeros(size(time));
for i = 2:length(time)
    fit(i) = fit(i-1) + (2*pi*kf*mt(i) *Ts);
end
Ac =5;
```

```
yt = Ac.*(cos(500*pi.*time+ fit));

%%
figure Name 'fi(t) in time domain'
plot(time, fit,"LineWidth",2)
title('fi(t) signal in time domain')
%xlim([-0.01,0.01])
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- fi(t) in time
domain'),'png')
fig_counter=fig_counter+ 1;

%%
%%Take fourier transform using fft
fs = 1/Ts;
yf = fft(fit)/fs;%take the fourşer transform using fft
figure Name 'Fi(f)'
n = length(fit);
%take the shift of the spectra of the fourier transform
fshift = (-n/2:n/2-1)*(fs/n);
fifshift = fftshift(yf);
plot(fshift,abs(fifshift))
xlim([-20,20])%limit the frequency x axis
% between -10 and 10 for best view
title('|Fi(f)|')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- Fi(f)'),'png')
fig_counter=fig_counter+ 1;

%%
figure Name 'y(t) in time domian'
plot(time, yt,"LineWidth",2)
title('y(t) signal in time domain')
%xlim([-0.01,0.01])
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- y(t) signal in time
domain'),'png')
fig_counter=fig_counter+ 1;
%%
figure Name 'y(t) in time domian from 3.95 to 4.05'
plot(time, yt,"LineWidth",2)
title('y(t) signal in time domain from 3.95 to 4.05')
xlim([3.95,4.05])
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- y(t) in time domian
from 3.95 to 4.05'),'png')
fig_counter=fig_counter+ 1;
%%
figure Name 'y(t) in time domian from 1.95 to 2.05'
plot(time, yt,"LineWidth",2)
title('y(t) signal in time domain from 1.95 to 2.05')
xlim([1.95,2.05])
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- y(t) in time domian
from 1.95 to 2.05'),'png')
```

```
fig_counter=fig_counter+ 1;

%%
%%Take fourier transform using fft
yf = fft(yt)/length(yt);%take the fourşer transform using fft
fs = 1/Ts;
figure Name 'Y(f)'
n = length(yt);
%take the shift of the spectra of the fourier transform
fshift = (-n/2:n/2-1)*(fs/n);
yfshift = fftshift(yf);
plot(fshift,abs(yfshift))
xlim([-1000,1000])%limit the frequency x axis
% between -10 and 10 for best view
title('|Y(f)|')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- Y(f)'),'png')
fig_counter=fig_counter+ 1;
%%
%Take the derivative of the y(t) signal and plot in time domain
ydt = diff(yt)/Ts;
figure Name 'y(t) derivative yd(t) in time domain'
plot(time(1:end-1), ydt,"LineWidth",2)
title('y(t) derivative in time domain')
%xlim([-0.01,0.01])
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- y(t) derivative yd(t)
in time domain'),'png')
fig_counter=fig_counter+ 1;
%%
%Take fourier transform using fft and plot in frequency domain
ytdf = fft(ydt)/length(ydt);%take the fourşer transform using fft
fs = 1/Ts;
figure Name '|Yd(f)| '
n = length(ydt);
%take the shift of the spectra of the fourier transform
fshift = (-n/2:n/2-1)*(fs/n);
ytdfshift = fftshift(ytdf);
plot(fshift,abs(ytdfshift))
xlim([-1000,1000])%limit the frequency x axis
% between -10 and 10 for best view
title('|Yd(f)|')
xlabel('Frequency (Hz)')
saveas(gcf,append(save_path,string(fig_counter),'- |Yd(f)|'),'png')
fig_counter=fig_counter+ 1;

%%
sinc = @(x) sin(pi.*x)./(pi.*x);
%%
fshift = -100:0.001:100;
figure Name 'M(f) from analitic solution'
mfanalitic = (2.*sinc(2.*fshift)).*(exp(-1i*2*pi.*fshift)-exp(-
1i*2*pi*3.*fshift));
plot(fshift,abs(mfanalitic))
xlim([-50,50])%limit the frequency x axis
title('|M(f)| from analitic')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
```

```
%saveas(gcf,append(save_path,string(fig_counter),'- |M(f)| from
analitic'),'png')
%fig_counter=fig_counter+ 1;


%%
C1 = 1000*10^-9;
R1 = 100*10^3;


%%
at = envelope_detector(time(1:end-1),ydt,C1,R1);
figure Name 'a(t)'
plot(time(1:end-1), at)
title('a(t) signal: envelope detector circuit')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- a(t) signal: envelope
detector circuit'),'png')
fig_counter=fig_counter+ 1;
%%
%calculate the fourier transform of a(t)
af = fft(at)/length(at);
figure Name '|A(f)|'
n = length(at);
fshift = (-n/2:n/2-1)*(fs/n);
afshift = fftshift(af);
plot(fshift,abs(afshift))
xlim([-5000,5000])
title('|A(f)|')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- |A(f)|'),'png')
fig_counter=fig_counter+ 1;
%%
figure Name '|E(f)|'
%to filter the ripples, a low pass filter is used
% because ripples have high frequency
fc = 20;
[et,ef] = lowPassFilterFrequencyDomain(afshift,fshift,fs,fc);
plot(fshift,abs(ef))
xlim([-50,50])
title('|E(f)|: the signal A(f) fitered in frequency domain')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- |E(f)|'),'png')
fig_counter=fig_counter+ 1;
%%
figure Name 'e(t)'
plot(time(1:end-1),et)
title('e(t): the signal a(t) fitered to remove ripples')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- e(t): the signal b(t)
fitered to remove ripples'),'png')
fig_counter=fig_counter+ 1;
%%
figure Name 'en(t)'
ent = et./(2*pi*kf*Ac);
plot(time(1:end-1),ent)
```

```
title('en(t): e(t) signal normalized')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- en(t): e(t) signal
normalized'),'png')
fig_counter=fig_counter+ 1;

%%
%Lets plot m(t) and e(t)

figure Name 'm(t) and en(t) signals'
hold on
plot(time,mt,"Marker","*",MarkerIndices=1:10:length(mt))
plot(time(1:end-1),ent)
title('m(t) and en(t) signals')
legend('m(t)','en(t)')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- m(t) and en(t)
signals'),'png')
fig_counter=fig_counter+ 1;
%%
%filter the dc signal of e(t)
zt = filterDC(ent);
figure Name 'z(t)'
plot(time(1:end-1),zt)
title('z(t): the signal en(t) fitered by capacitor to remove dc signal')
xlabel('Time(sec)')
saveas(gcf,append(save_path,string(fig_counter),'- z(t): the signal en(t)
fitered'),'png')
fig_counter=fig_counter+ 1;
%%
%calculate the fourier transform of z(t)
zf = fft(zt)/fs;
figure Name '|Z(f)|'
n = length(zt);
fshift = (-n/2:n/2-1)*(fs/n);
zfshift = fftshift(zf);
plot(fshift,abs(zfshift))
xlim([-50,50])
title('|Z(f)|')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
saveas(gcf,append(save_path,string(fig_counter),'- |Z(f)|'),'png')
fig_counter=fig_counter+ 1;

%%
figure Name 'm(t) and z(t)'
hold on
plot(time,mt,"Marker","*",MarkerIndices=1:10000:length(mt))
plot(time(1:end-1),zt)
title('m(t) and z(t) signals')
legend('m(t)','z(t)')
xlabel('Time(sec)')
ylabel('Amplitude')
saveas(gcf,append(save_path,string(fig_counter),'- m(t) and z(t)
signals'),'png')
fig_counter=fig_counter+ 1;

%%
```

```
function [at] = envelope_detector(time,yt,C1,R1)
   at = zeros(size(yt));
   at(1)= yt(1);
   i0 = 1;
   for i = 2:length(time)
        if yt(i)>0 && yt(i)>at(i-1)
            if yt(i) >= yt(i- 1)
                at(i) = yt(i);
                i0 = i;
            else
                at(i) = yt(i0).* exp(-(time(i)- time(i0))/(R1.*C1));
            end
        else
            at(i) = yt(i0).* exp(-(time(i)- time(i0))/(R1.*C1));
        end
    end


end

%%
function [et,ef] = lowPassFilterFrequencyDomain(bf,f,fs,fc)
% Compute the magnitude response of the filter
   H = zeros(size(f));
    for i = 1:length(f)
        if abs(f(i)) <= fc
            H(i) = 1;
        else
            H(i) = 0;
        end
    end
    % Apply the filter to the frequency domain signal
    ef = bf .* H;
    % Obtain the time-domain signal using the inverse Fourier transform
    et = ifft(ifftshift(ef)).*length(bf);
end
%%

function [yt] = square_wave(xt)
% return a square wave with amplitude equal to 1 and width 1
    yt = zeros(size(xt));
    for i = 1:length(xt)
        if abs(xt(i)) >= -1/2 && abs(xt(i)) <=  1/2
            yt(i) = 1;
        else
            yt(i) = 0;
        end
    end
end

%%
% function to filter DC signal of e(t) signal by taking the average
%  of the signal and subtracting it from the signal

function [zt] = filterDC(et)
    zt = et - mean(et);
end
```